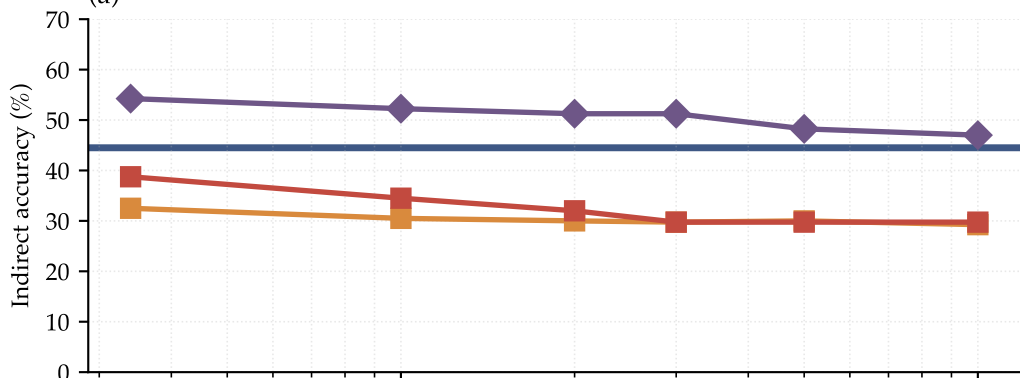
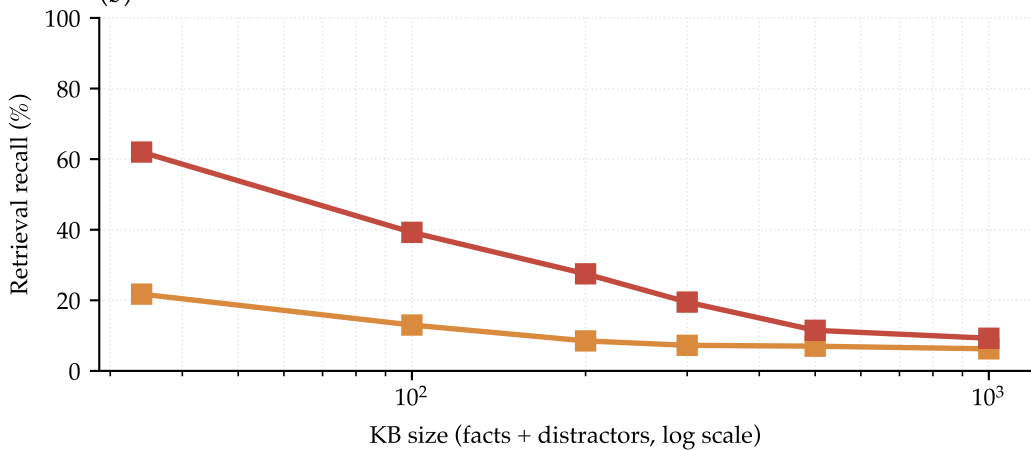


(a)



(b)



KB size (facts + distractors, log scale)

Mini-Engram + RAG top-1

RAG top-3 + shared reasoning LoRA

Mini-Engram + RAG top-3

Engram rows + shared reasoning LoRA (no retrieval) = 44%